The Pyramid: addendum

Errors found in the 2022 version of The Pyramid (AWEC 2021), as of 2026-11-07

The calculations of the paper were redone in samara (**docs/the_pyramid/pyramid.samara**, checked by **checks.samara**), and the Julia listings of the paper were run again. This addendum lists what differs from the 2022 text. Each item says how it was checked. The 2022 PDF is not changed.

1. Table of shaft geometries: two values truncated

The table gives $\Lambda = r_0 / \sqrt{(l^2 - r_0^2 - r_1^2)}$, the value at a twist of 90°, which is the formula the other rows follow. Shafts C and D are cut off, not rounded:

Shaft	l	r0	r1	In the paper	Computed	Rounded
C	200 m	20 m	50 m	0.103	0.10383	0.104
D	200 m	10 m	100 m	0.057	0.05783	0.058

Rows A, B, E and F (0.052, 0.116, 0.025, 0.237) are right to the digits given. Checked by computing all six rows.

2. The power plots are drawn at 60°, not 30°

The text says the plots use an elevation angle of 30° “unless otherwise stated”. The Julia listings in the paper set **elevation_angle = pi / 3**, which is 60°, in the plots of power against Λ for wind speed, glide number and the single curve. All of them also cut the Λ axis at 0.04 (**xlims = (0, 0.04)**). The “numerical solution” column of the glide-number table agrees with these plots, not with a run at 30° (see below). Running the paper's own functions (**lambda_power_sweep**, 12 m/s wind, C_L = 1.5, wing area 3 m²) and taking the Λ of the maximum power gives:

C_L/C_D	Paper, numeric	Computed at 30°	Computed at 60°	Equation (27)
5	> 0.040	0.1026	0.1315	0.100
10	0.040	0.0503	0.0549	0.050
15	0.035	0.0334	0.0349	0.033
20	0.025	0.0250	0.0256	0.025

At 30° equation (27) agrees with the numerical optimum to within 3 % for all four glide numbers, better than the printed column suggests. The entries for 5 and 10 are the edge of the axis, not the optimum, and the 0.035 for 15 is the 60° value (0.0349), not the 30° one. The statement that the optimal Λ depends little on the elevation angle holds for high glide numbers (about 2 % apart at C_L/C_D = 20) but not for low ones (28 % apart at 5). Checked by running the listing at both angles.

3. The “maximum Λ” line is the value at 90°, not the maximum

The paper takes the maximum of the moment at θ ≈ 90° and leaves the exact maximum out. The dashed red line in the shaft-length plots is “an absolute upper limit for that geometry”, but it is drawn as $r_0 / \sqrt{(l^2 - r_0^2 - r_1^2)}$, the 90° value. The exact maximum over the twist is

$\Lambda_{max} = (\sqrt{(l^2 - (r_1 - r_0)^2)} - \sqrt{(l^2 - (r_1 + r_0)^2)}) / (2\, r_1)$, which goes to r_0 / l for a long shaft.

For r_0 = 20 m, r_1 = 50 m and l = 100 m (shaft F) the 90° value is 0.2374 and the maximum is 0.2398, at a twist of 98.3° (1.0 % higher). The line therefore lies a little below the true limit. Checked by a search over the twist; the closed form agrees with it to 8 digits.

4. Julia listing of the two-section shaft: a call with a missing argument

In **find_possible_buckling_theta** the call **lambda(r0, r2, l0 + l1)** has three arguments, but **lambda** takes four (the last is the twist). Julia raises a **MethodError**, and the surrounding **try / catch** turns it into **(false, NaN)**. The buckling case is therefore never used, and **max_lambda** always takes the sampled maximum. The corrected call is **lambda(r0, r2, l0 + l1, th)**. Confirmed by calling it. How far the plot of

the maximum Λ against the section split point changes has not been measured.

Not checked

The tether drag formulas and the shaft-length results of the section “Tether drag” rest on the companion note «A simplified drag estimate for a tether with a belly», which was not part of this check; the closed form for the drag in the Julia listing (**belly_drag_shaft**) was not compared against it. The figures were not redrawn, and the kite simulations with TRPTSim were not repeated.

Additions

Beside the corrections, the samara version gives two results the paper does not have. With the elevation angle θ_e kept symbolic, the optimal power equals Loyd's crosswind power with the wind w replaced by w cos θ_e, and the optimal Λ is 1 / (2 G) for every θ_e. The exact maximum of Λ over the twist is the formula of item 3.